

Telecommunications Customer Dataset - Complete Business Analysis Documentation

Executive Summary

This dataset contains 1,000+ telecommunications customer records focused on churn prediction. The telecom industry typically faces 15-25% annual churn rates, making customer retention a critical business priority where each lost customer represents significant revenue loss and high acquisition costs.

Business Context & Industry Overview

- Industry:** Telecommunications/Internet Service Provider
- Primary Business Challenge:** Customer churn prediction and retention
- Data Purpose:** Customer analytics, revenue optimization, and strategic decision-making
- Key Business Driver:** Customer retention is 5-25x more cost-effective than acquisition

Dataset Structure & Metrics

Category	Columns	Business Purpose
Customer ID	1	Account management
Demographics	4	Market segmentation
Service Portfolio	9	Revenue optimization
Contract & Billing	3	Risk assessment
Financial Metrics	2	Revenue analysis
Target Variable	1	Churn prediction
Total	21	Comprehensive customer view

Detailed Column Documentation

Customer Identification

Column	Data Type	Description	Business Impact
customerID	String	Unique customer identifier	Primary key for customer tracking and CRM integration

Demographics Portfolio

Column	Data Type	Values	Business Impact	Formula/Calculation
gender	Categorical	Male,	Market segmentation for	Gender Mix = Count(Male)/Total * 100

Column	Data Type	Values	Business Impact	Formula/Calculation
		Female	targeted campaigns	
SeniorCitizen	Binary	0=No, 1=Yes	Senior-specific pricing and service strategies	Senior Rate = Sum(SeniorCitizen)/Total * 100
Partner	Categorical	Yes, No	Household stability and family plan targeting	Partner Rate = Count(Partner='Yes')/Total * 100
Dependents	Categorical	Yes, No	Family service needs assessment	Dependent Rate = Count(Dependents='Yes')/Total * 100

Account & Tenure Information

Column	Data Type	Range	Business Impact	Key Formulas
tenure	Numeric	0-72 months	Customer lifetime value and loyalty measurement	Avg Tenure = Mean(tenure) Churn by Tenure = Churn Rate per tenure group

Service Portfolio Matrix

Column	Data Type	Values	Business Impact	Penetration Formula
PhoneService	Categorical	Yes, No	Core service adoption tracking	Phone Penetration = Count(PhoneService='Yes')/Total
MultipleLines	Categorical	Yes, No, No phone service	Service complexity and upsell opportunities	Multiple Lines Rate = Count(MultipleLines='Yes')/Count(PhoneService='Yes')
InternetService	Categorical	DSL, Fiber optic, No	Primary revenue driver and technology strategy	Internet Penetration = Count(InternetService≠'No')/Total Fiber Mix = Count('Fiber')/Count(Internet='Yes')
OnlineSecurity	Categorical	Yes, No, No internet service	Value-added service attachment	Security Attach Rate = Count(OnlineSecurity='Yes')/Count(Internet='Yes')
OnlineBackup	Categorical	Yes, No, No internet service	Data protection service revenue	Backup Attach Rate = Count(OnlineBackup='Yes')/Count(Internet='Yes')

Column	Data Type	Values	Business Impact	Penetration Formula
DeviceProtection	Categorical	Yes, No, No internet service	Hardware protection revenue stream	Device Protection Rate = Count(DeviceProtection='Yes')/Count(Internet='Yes')
TechSupport	Categorical	Yes, No, No internet service	Customer satisfaction and premium tier	Tech Support Rate = Count(TechSupport='Yes')/Count(Internet='Yes')
StreamingTV	Categorical	Yes, No, No internet service	Entertainment bundle penetration	TV Streaming Rate = Count(StreamingTV='Yes')/Count(Internet='Yes')
StreamingMovies	Categorical	Yes, No, No internet service	Content service completion	Movie Streaming Rate = Count(StreamingMovies='Yes')/Count(Internet='Yes')

Contract & Payment Analytics

Column	Data Type	Values	Business Impact	Risk Formulas
Contract	Categorical	Month-to-month, One year, Two year	Commitment level and churn risk assessment	Contract Mix = Count(each contract type)/Total Risk Score = Higher for month-to-month
PaperlessBilling	Categorical	Yes, No	Operational efficiency and cost reduction	Paperless Rate = Count(PaperlessBilling='Yes')/Total
PaymentMethod	Categorical	Electronic check, Mailed check, Bank transfer (automatic), Credit card (automatic)	Payment reliability and operational costs	Auto Payment Rate = Count(Automatic methods)/Total

Financial Performance Metrics

Column	Data Type	Range	Business Impact	Key Calculations
MonthlyCharges	Numeric	\$18.25-\$118.75	Revenue per	ARPU = Mean(MonthlyCharges)

Column	Data Type	Range	Business Impact	Key Calculations
			customer and pricing strategy	 Revenue Quartiles = Percentile analysis
TotalCharges	Numeric	Varies (some missing)	Customer lifetime value calculation	CLV = Mean(TotalCharges) Revenue per Tenure = TotalCharges/tenure

Target Variable

Column	Data Type	Values	Business Impact	Core Metrics
Churn	Binary	Yes, No	Primary business outcome for retention	Churn Rate = Count(Churn='Yes')/Total * 100 Retention Rate = 100 - Churn Rate

Critical Business KPIs & Formulas

Primary Customer Metrics

KPI	Formula	Industry Benchmark	Business Significance
Overall Churn Rate	(Churned Customers / Total Customers) × 100	15-25% annually	Core retention performance
Monthly Churn Rate	Churn Rate / 12	1.25-2.1% monthly	Short-term trend monitoring
Customer Lifetime Value	Average(TotalCharges/tenure)	Varies by segment	Investment prioritization
Average Revenue Per User	Mean(MonthlyCharges)	\$50-80 typical	Revenue efficiency

Service Performance Metrics

KPI	Formula	Target Range	Strategic Impact
Internet Penetration	(Internet Customers / Total) × 100	80-95%	Market saturation
Fiber Adoption	(Fiber Customers / Internet Customers) × 100	40-60%	Technology transition
Service Bundle Rate	Average services per customer	3-5 services	Revenue per customer
Add-on Attachment	(Add-on services / Internet customers) × 100	60-80%	Upsell effectiveness

Financial KPIs

KPI	Formula	Business Driver
Monthly Recurring Revenue	Sum(MonthlyCharges for active customers)	Predictable revenue stream
Revenue at Risk	Sum(MonthlyCharges for high churn risk customers)	Retention campaign prioritization
Average Revenue per Service	Total Revenue / Total Service Subscriptions	Service profitability

Risk Assessment Metrics

Risk Factor	Calculation	Risk Level
Month-to-month %	(Month-to-month contracts / Total) × 100	High risk: >60%
New Customer Churn	Churn rate for tenure < 12 months	Critical: >30%
Electronic Check Risk	Churn rate for electronic check users	Monitor: >25%
Single Service Risk	Churn rate for customers with 1 service	Concern: >40%

Customer Segmentation Framework

Value-Based Segments

Segment	MonthlyCharges Range	Characteristics	Strategy
High Value	\$80+	Premium services, multiple add-ons	Premium retention programs
Medium Value	\$40-79	Core services with some add-ons	Upsell opportunities
Low Value	<\$40	Basic services only	Cost optimization

Tenure-Based Lifecycle

Stage	Tenure Range	Churn Risk	Focus Area
New	0-6 months	Highest	Onboarding excellence
Growing	6-24 months	Medium	Service expansion
Mature	24+ months	Lowest	Loyalty rewards

Service-Based Segments

Segment	Service Mix	Revenue Impact	Retention Strategy
Bundle Champions	4+ services	Highest ARPU	VIP treatment
Core Users	Internet + 1-2 add-ons	Moderate ARPU	Expansion campaigns

Segment	Service Mix	Revenue Impact	Retention Strategy
Basic Users	Phone or Internet only	Lowest ARPU	Bundle incentives

Data Quality Assessment

Column	Issue Type	Impact	Recommendation
TotalCharges	Blank values	CLV calculations affected	Investigate if new customers (tenure=0)

Data Anomalies to Investigate

Anomaly	Description	Business Implication
Tenure = 0	New customers with no history	May indicate data collection issues
Streaming without Internet	Logical inconsistency	Data validation needed

Strategic Analysis Framework

1. Churn Analysis Priority Matrix

Risk Level	Tenure	Contract Type	Action Required
Critical	<6 months	Month-to-month	Immediate intervention
High	6-12 months	Month-to-month	Retention campaign
Medium	12+ months	Month-to-month	Contract upgrade offer
Low	Any	Annual/2-year	Maintain satisfaction

2. Revenue Optimization Opportunities

Opportunity	Target Segment	Revenue Impact	Implementation
Fiber Upsell	DSL customers	\$10-20/month increase	Technology migration campaign
Service Bundling	Single service users	\$15-30/month increase	Bundle promotions
Contract Extension	Month-to-month	15-25% churn reduction	Incentive pricing

3. Customer Journey Analytics

Journey Stage	Key Metrics	Success Indicators
Acquisition	New customer rate	>10% monthly growth
Onboarding	30-day activation	>90% success rate

Journey Stage	Key Metrics	Success Indicators
Expansion	Service additions	>2 services per customer
Retention	Churn prevention	<20% annual churn

Recommended Analytics Roadmap

Phase 1: Descriptive Analytics (Week 1-2)

- Customer distribution analysis
- Service adoption patterns
- Revenue concentration analysis
- Churn rate by segment

Phase 2: Diagnostic Analytics (Week 3-4)

- Churn driver identification
- Service bundle impact analysis
- Payment method correlation
- Tenure-based behavior patterns

Phase 3: Predictive Analytics (Week 5-8)

- Churn probability modeling
- Customer lifetime value prediction
- Service upgrade propensity
- Revenue forecasting

Phase 4: Prescriptive Analytics (Week 9-12)

- Retention campaign optimization
- Pricing strategy recommendations
- Service bundle optimization
- Customer journey personalization

This comprehensive framework provides the foundation for data-driven decision making in telecommunications customer management and strategic planning.